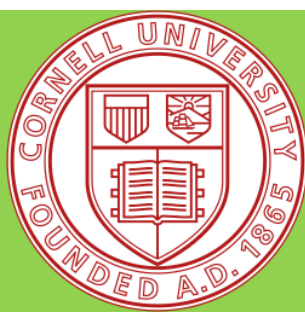




Investigating the Limitations of Transformers with Arithmetic Tasks

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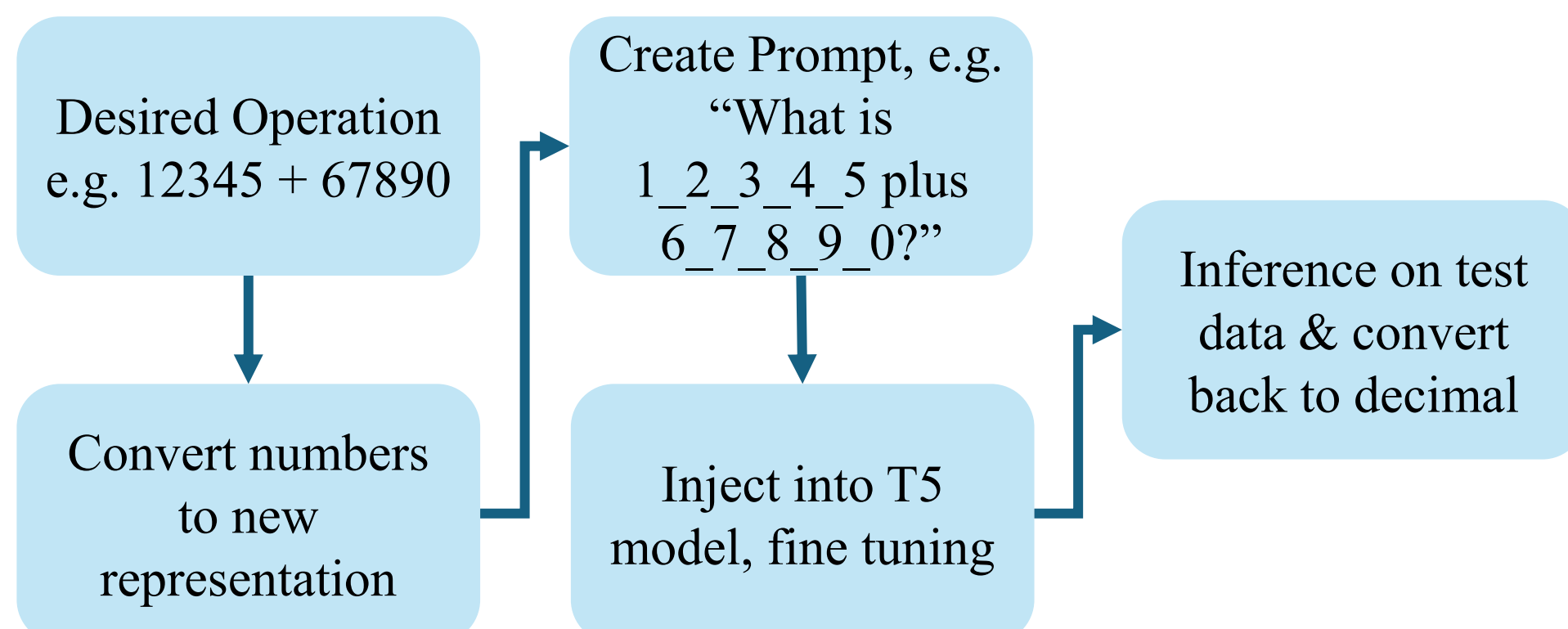
Introduction

- **LLMs are notorious for being unable to perform arithmetic.** Even straightforward decimal inputs often lead to incorrect or nonsensical outputs.
- **Goal:** Investigate the effect of number representation on arithmetic performance.
- **Reproduce results** on the addition tasks from the original paper and compare accuracy statistics across the various representations.

Methodology

7 number representations. Structured template: **two numbers and an operator** → output

- **Formats:** decimal, character-based, fixed-character, underscore, word-based, 10-based, 10E-based
- **Data:** we generated our own datasets containing 22,000 arithmetic problems.
- **Base Model:** T5 (sequence-to-sequence)
- **Parameters:** Learning rate=0.00005, weight decay=0.00005, epochs=20



Results and Evaluation

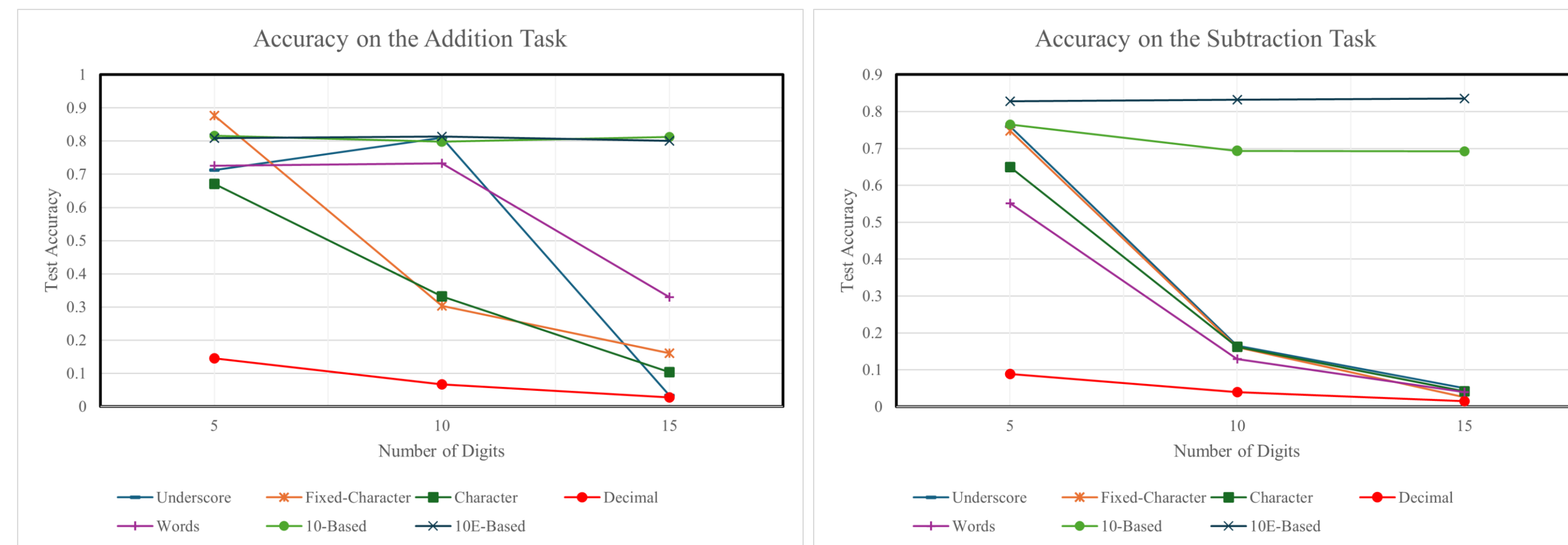


Figure 1: Accuracy of various representations on the addition task

Figure 2: Accuracy of various representations on the subtraction task

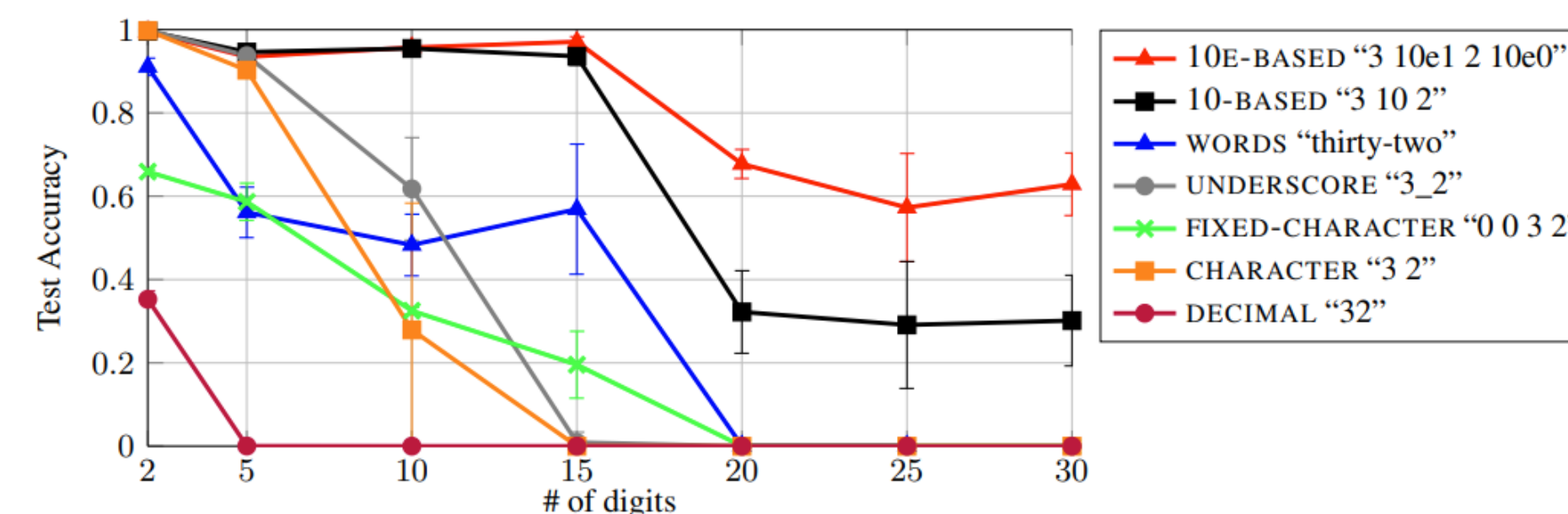


Figure 3: Test Accuracy from the original paper

- Results match prior work (Fig. 3): 10-based and 10E-based representations perform best, especially for larger digit lengths.
- Decimal representation performs worst, with rapid accuracy degradation as digit length increases (Fig. 1).
- Subtraction results (Fig. 2) show the same overall trend.

Conclusion & Discussion

# of digits	Under-score	Fixed-Char.	Char.	Deci-mal	Words	10-based	10E-based
15	0.033	0.161	0.104	0.028	0.329	0.812	0.8

Conclusion:

- Our results on the addition tasks are very similar to that of the original paper.
- Using 10 and 10E-based surface representation leads to vastly significant improvements in accuracy.
- For other representations, accuracy drops as number of digits increases.

Future Work:

- Study arithmetic in non-base-10 systems, such as hexadecimal.
- Compare pretrained vs. base-specific trained models.
- Extend experiments beyond bases 2, 3, 10, and 19.

Acknowledgements

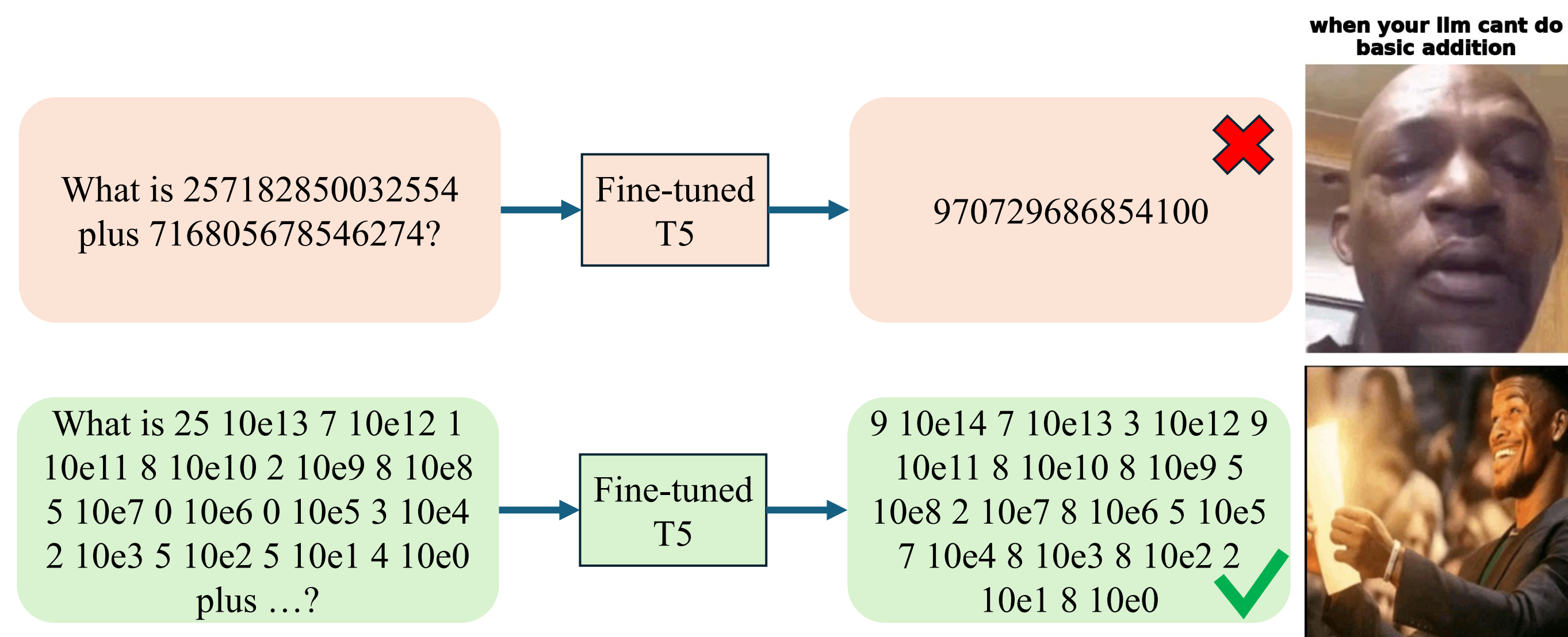
Original Paper: "Investigating the Limitations of Transformers with Simple Arithmetic Tasks"
Rodrigo Nogueira, Zhiying Jiang & Jimmy Lin
April 12, 2021

<https://arxiv.org/pdf/2102.13019>

T5 Model: "Exploring the limits of transfer learning with a unified text-to-text transformer"
Raffel et al.
Journal of Machine Learning Research, 2020

All charts and diagrams created by the authors of this poster.

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A visualization of examples taken from our test dataset